Cauchy-characteristic matching for the Z4c evolution system: method

z4c-CMM verified research loop[1]

[1] *generated from the mission-2 knowledge ledger (`knowledge-database/paper_z4c-CCM/`); every equation and number below is traceable to a verifier output under `results/numerical/`*

(Dated: June 12, 2026)

I. METHOD

Black equations are transcribed from Refs. [1–3]; dark blue equations constitute the new Z4c-CCM formulation. Verification labels [Vn] refer to Table I; dynamical evidence is in Table II.

A. Boundary frame and Z4c variables

$$ds^2 = -\alpha^2\, dt^2 + \gamma_{ij}\,(dx^i + \beta^i dt)(dx^j + \beta^j dt), \qquad n^\mu = \alpha^{-1}\,(1,\, -\beta^i), \tag{1}$$

$$s_i = \frac{\partial_i r}{\sqrt{\gamma^{rr}}}, \qquad s^i = \frac{\gamma^{ir}}{\sqrt{\gamma^{rr}}}, \qquad m^A = \tfrac{1}{\sqrt{2}}\big(e_2 + i\, e_3\big)^A, \tag{2}$$

$$\mathring{l}^\mu = \tfrac{1}{\sqrt{2}}\,(n^\mu + s^\mu), \qquad \mathring{k}^\mu = \tfrac{1}{\sqrt{2}}\,(n^\mu - s^\mu) \qquad [1]. \tag{3}$$

Z4c variables [1]:

$$\chi = (\det\gamma)^{-1/3}, \quad \tilde\gamma_{ij} = \chi\,\gamma_{ij}, \quad K = \hat K + 2\Theta, \quad \tilde A_{ij} = \chi\Big(K_{ij} - \tfrac{1}{3}\gamma_{ij}K\Big), \tag{4}$$

$$\partial_t\chi = \tfrac{2}{3}\,\chi\,\big[\alpha(\hat K + 2\Theta) - D_i\beta^i\big], \tag{5}$$

$$\partial_t\tilde\gamma_{ij} = -2\alpha\tilde A_{ij} + \beta^k\tilde\gamma_{ij,k} + 2\tilde\gamma_{k(i}\,\beta^k_{\ ,j)} - \tfrac{2}{3}\,\tilde\gamma_{ij}\,\beta^k_{\ ,k} \qquad [1]. \tag{6}$$

B. Worldtube data map (node N1)

$$g_{00} = -\alpha^2 + \gamma_{ij}\beta^i\beta^j, \qquad g_{0i} = \gamma_{ij}\beta^j, \qquad g_{ij} = \gamma_{ij}; \qquad g^{00} = -\alpha^{-2}, \quad g^{0i} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i\beta^j}{\alpha^2}. \tag{7}$$

ADM kinematics requires $\partial_t\gamma_{ij} = -2\alpha K_{ij} + \beta^k\partial_k\gamma_{ij} + \gamma_{kj}\partial_i\beta^k + \gamma_{ik}\partial_j\beta^k$. From $\gamma_{ij} = \chi^{-1}\tilde\gamma_{ij}$,

$$\partial_t\gamma_{ij} = \chi^{-1}\partial_t\tilde\gamma_{ij} - \chi^{-2}\tilde\gamma_{ij}\,\partial_t\chi, \tag{8}$$

and inserting (5)–(6) with $K_{ij} = \chi^{-1}\big(\tilde A_{ij} + \tfrac{1}{3}\tilde\gamma_{ij}K\big)$, $\beta^k\partial_k\gamma_{ij} = \chi^{-1}\beta^k\partial_k\tilde\gamma_{ij} - \gamma_{ij}\,\beta^k\partial_k\ln\chi$,

$$\big[\partial_t\gamma_{ij}\big]_{\rm Z4c} - \big[\partial_t\gamma_{ij}\big]_{\rm ADM} = \gamma_{ij}\Big[\tfrac{2}{3}\,D_k\beta^k - \tfrac{2}{3}\,\partial_k\beta^k + \beta^k\partial_k\ln\chi\Big]. \tag{9}$$

$$\boxed{D_i\beta^i = \partial_k\beta^k - \tfrac{3}{2}\,\beta^k\,\partial_k\ln\chi} \qquad \text{[V3: closure exact; naive } D = \partial_k\beta^k \text{ leaves residual } +1\cdot\gamma_{ij}\beta^k\partial_k\ln\chi]. \tag{10}$$

C. Tetrad correspondence and Type-II boost (node N2)

Outgoing null label u with $u|_{\partial\Omega} = t$: $\partial_\mu u = (1,\, u_r,\, 0,\, 0)$,

$$g^{tt} + 2g^{tr}u_r + g^{rr}u_r^2 = 0 \quad\Longrightarrow\quad u_r = \frac{-g^{tr} - \sqrt{(g^{tr})^2 - g^{tt}g^{rr}}}{g^{rr}}, \qquad u_r\big|_{\text{flat}} = -1. \tag{11}$$

Characteristic outgoing null vector [2, 3]:

$$\hat{l}^{\,\mu} = -e^{2\hat{\beta}}\, g^{\mu\nu}\partial_\nu u. \tag{12}$$

For arbitrary $(\alpha, \beta^i, \gamma_{ij})$,

$$\mathring{l}^{\mu} = \frac{\sqrt{2}}{2}\left(\alpha - \gamma_{ij}\beta^i s^j\right) e^{-2\hat{\beta}}\, \hat{l}^{\,\mu} \qquad \text{[V1: exact at 12 rational points; GPU residual } 9.3\times10^{-16}\text{]}, \tag{13}$$

$$A = \left(\alpha - \gamma_{ij}\beta^i s^j\right) e^{-2\hat{\beta}}, \qquad \psi_0' = A^2\,\psi_0, \tag{14}$$

i.e. the Choice-2 boost of Ref. [3] holds verbatim on the Z4c frame (3): no evolution-system variable enters (13).

ψ_0 on the characteristic grid [2, 3]:

$$\psi_0 = \Big(\frac{r\partial_r\beta - 1}{4Kr}\Big)\Big[(1{+}K)\partial_r J - \frac{J^2\partial_r\bar{J}}{1{+}K}\Big] + \frac{J(1{+}K^2)\,\partial_r J\,\partial_r\bar{J}}{8K^3} + \frac{1}{8K}\Big[\frac{J^2\partial_r^2\bar{J}}{1{+}K} - (1{+}K)\partial_r^2 J\Big] - \frac{J\bar{J}^2(\partial_r J)^2 + J^3(\partial_r\bar{J})^2}{16K^3}. \tag{15}$$

D. Physical-mode dictionary (node N3, exact)

Electric/magnetic Weyl decomposition w.r.t. n^μ (vacuum):

$$E_{\mu\nu} = C_{\mu\alpha\nu\beta}\, n^\alpha n^\beta, \qquad B_{\mu\nu} = \tfrac{1}{2}\,\epsilon_{\mu\alpha}{}^{\gamma\delta}C_{\gamma\delta\nu\beta}\, n^\alpha n^\beta, \qquad \epsilon_{\mu\nu\rho\sigma} = \sqrt{-g}\,[\mu\nu\rho\sigma]. \tag{16}$$

EXACT identity (pure Weyl algebra, generic 10-component tensor; no perturbation, no field equations) [V2a]:

$$\left(E_{\mu\nu} + \epsilon_\mu{}^{\kappa\lambda}s_\kappa B_{\lambda\nu}\right) m^\mu m^\nu \;=\; -\,\psi_0, \qquad \left(E_{\mu\nu} - \epsilon_\mu{}^{\kappa\lambda}s_\kappa B_{\lambda\nu}\right) m^\mu m^\nu \;=\; -\,\bar{\psi}_4, \tag{17}$$

with $\psi_0 = -C_{abcd}\, l^a m^b l^c m^d$ on the frame (3); dyad completeness ($U_{AB} = U(m,m)\,\bar{m}_A\bar{m}_B + \text{c.c.}$ for tangential trace-free U) gives the tensor form of the incoming Weyl object

$$U^-_{AB} \;\equiv\; \mathrm{TT}_2\big[E + \epsilon(s)B\big]_{AB} \;=\; -\big(\psi_0\,\bar{m}_A\bar{m}_B + \bar{\psi}_0\, m_A m_B\big). \tag{18}$$

Realization from the Z4c state (Gauss–Codazzi, on-shell vacuum; γ_{ij}, K_{ij} from (4)) [V2c]:

$$E_{ij} = R_{ij}[\gamma] + K\,K_{ij} - K_{ik}K^k{}_j, \qquad B_{ij} = \epsilon_i{}^{kl}\, D_k K_{lj}\big|_{\text{sym}}, \qquad \epsilon_{ikl} = \sqrt{\det\gamma}\,[ikl]. \tag{19}$$

Nonperturbative verification (Kerr–Schild Schwarzschild, autodiff, both routes; Table I V2c): $|E^{3+1} - E^{4D}|$ rel. 7.7×10^{-17}, $|B^{3+1} - B^{4D}|$ rel. 3.9×10^{-17}, identity residual 8.5×10^{-16}. Linearized on-shell flat limit (superseded defining form, now a corollary [V2b]): for TT $h_{AB}(t, x_s)$, $\psi_0 = \frac{1}{4}(\partial_t + \partial_s)^2 h_{mm}$ and $U^-_{AB} \to -\frac{1}{4}(\partial_t + \partial_s)^2\gamma^{\mathrm{TF}}_{AB}$, i.e. $(\partial_t + \partial_s)^2\gamma^{\mathrm{TF}}_{AB} = 4(\psi_0\bar{m}_A\bar{m}_B + \text{c.c.})$.

E. The Z4c-CCM boundary conditions (node N6)

Ten incoming modes at the worldtube; $\mathring{X}^\mu_v = (n^\mu + \sqrt{v}\, s^\mu)/\sqrt{2}$ at sector speed v; $L \geq 1$:

$$\text{physical (2):} \qquad U^-_{AB} \mathrel{\hat{=}} -\Big(\psi_0'\,\bar{m}_A\bar{m}_B + \bar{\psi}_0'\, m_A m_B\Big), \qquad \psi_0' = A^2\,\psi_0^{\mathrm{CCE}} \quad \text{(exact, (18)–(19))}, \tag{20}$$

$$\text{constraint (4):}\qquad \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}\Theta \mathrel{\hat{=}} 0,\qquad \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}Z_s \mathrel{\hat{=}} 0,\qquad \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}Z_A \mathrel{\hat{=}} 0, \tag{21}$$

$$\text{gauge (4):}\qquad \big(r^2\,\mathring{m}\cdot\partial\big)^{L+1}\alpha \mathrel{\hat{=}} 0,\qquad \big(r^2\,\mathring{k}_{\nu_s}\cdot\partial\big)^{L+1}\beta_s \mathrel{\hat{=}} 0,\qquad \big(r^2\,\mathring{j}\cdot\partial\big)^{L+1}\beta_A \mathrel{\hat{=}} 0. \tag{22}$$

Bjørhus form of (20) (the correction drives the incoming Weyl object to its CCM value, the Z4c analog of Ref. [3]):

$$U^-_{AB} - U^-_{AB}\big|_{\rm BC} \to 0,\qquad U^-_{AB}\big|_{\rm BC} = -\Big(\psi_0'\,\bar m_A \bar m_B + \bar\psi_0'\,m_A m_B\Big);\qquad \psi_0^{\rm CCE}\to 0 \implies \text{CPBC scheme of Ref. [1]}\quad \text{[V6]}. \tag{23}$$

Equations (21)–(22) are the order-L conditions of Ref. [1]; (22) replaces the Sommerfeld gauge placeholder of Ref. [3].

F. Sector orthogonality (nodes N4, N5)

For arbitrary TT data $h_{AB}(t,x_s)$ [V4]:

$$\mathcal{H} = \partial_i\partial_j h_{ij} - \partial^2 h^k{}_k = 0,\qquad \mathcal{M}_i = \partial_j \dot h_{ij} - \partial_i \dot h^j{}_j = 0,\qquad \gamma^{ij}\partial_t h_{ij} = 0,\qquad \tilde\Gamma^i = -\partial_j \tilde h^{ij} = 0; \tag{24}$$

negative controls: $h_{x_s y} = G(t,x_s) \Rightarrow \mathcal{M}_y = -\tfrac12\,\partial_{x_s}\dot G \neq 0$; $h_{yy} = h_{zz} = G \Rightarrow \mathcal{H} = -2\,\partial^2_{x_s} G \neq 0$.

Hence the datum (20) sources neither (21) nor (22) at linear order: the three sectors of (20)–(22) commute.

On the boundary wave model of Ref. [1], $\big[\partial_t^2 - 2\mathring\beta\,\partial_t\partial_x - (1-\mathring\beta^2)\partial_x^2\big]u = 0$, plane-wave reflection [V5]:

$$R_{\rm Sommerfeld} = -\frac{\mathring\beta\,(1-\mathring\beta)}{(1+\mathring\beta)(2-\mathring\beta)} \neq 0,\qquad R_{(\partial_t + c_+\partial_x)} = 0\quad \text{(identically)}. \tag{25}$$

G. Frozen-coefficient boundary stability (node N7)

Laplace–Fourier ansatz $u = \tilde u\, e^{st+\lambda x}$, $\Re s > 0$ [V7]:

$$\lambda_+ = \frac{s}{1+\mathring\beta}\ \text{(admissible)},\qquad \lambda_- = -\frac{s}{1-\mathring\beta},\qquad D_L(s) = \Big(s + (1+\mathring\beta)\,\lambda_+\Big)^{L+1} = (2s)^{L+1},\qquad \frac{|D_L(s)|}{|s|^{L+1}} = 2^{L+1}. \tag{26}$$

D_L contains no datum symbol: the injection (20) modifies only the inhomogeneity, so the boundary-stability results of Ref. [1] (constraint sector: all L; gauge+metric sector: spherical reduction, asymptotically harmonic shift) transfer verbatim with $\|u\| \le C\,\|\psi_0'\text{-datum}\|$.

H. Numerical evidence

All runs: JAX float64, 4× NVIDIA A100-80GB (`CUDA_VISIBLE_DEVICES=0,1,2,3`), wall clock ≤ 23 s each (budget 600 s); outputs under `results/numerical/`.

Table I. Symbolic and sampling verification of (10)–(26). "Exact" = sympy identity on arbitrary functions or exact rationals (zero float error). Files: `n2_boost_check.txt`, `n3_exact_check.txt`, `n3_dictionary_check.txt`, `n1_varmap_check.txt`, `n4_cpbc_compat_check.txt`, `n5_gauge_check.txt`, `n6_composite_check.txt`, `n7_lf_sketch_check.txt`.

	verified statement	exact part	GPU sweep (4 devices)
V1	boost identity (13), factor $\sqrt{2}/2$	12/12 rational points	9.34×10^{-16} over 4,194,304 samples
V2a	EXACT identity (17)–(18), $(\sigma,\kappa)=(1,-1)$, $\bar\psi_4$ partner	generic Weyl, all 10 dof	—
V2b	linearized on-shell limit (corollary)	arbitrary F_1, F_2	0.0 over 4,194,304 modes (prior form)
V2c	Gauss–Codazzi realization (19), Schwarzschild	Codazzi sign +1 derived	E: 7.7×10^{-17}, B: 3.9×10^{-17}, identity 8.5
V3	maps (7), closure (10)	full-symbol identities	round trip 4.10×10^{-16}, gg^{-1} 8.88×10^{-16}
V4	orthogonality (24) + controls	identities and non-vanishing controls	TT residual 0.0 vs control 12.3
V5	reflection (25)	exact R algebra	$\lvert R_{\rm Som}\rvert \ge 0.023$ on $\mathring\beta \in [0.05, 0.6]$; $R=0$ ex
V6	10-mode count, $\psi_0\to0$ reduction, wire-through (exact targets)	structural	reduction 0.0; chain residual 1.47×10^{-16}
V7	Kreiss condition (26), $L=0..3$	$D_L=(2s)^{L+1}$ exact	ratio deviation 3.2×10^{-14} (4,194,304 sam

Table II. Dynamical model evolution (6th-order interior stencils with a GKS-stable mixed closure — 4th-order Fornberg rows at three edge nodes; fully one-sided 6th-order closures are RK4-unstable (error ledger) — RK4, CFL 0.2, shift $\mathring{\beta} = 0.3$, outgoing Gaussian pulse onto the boundary; one configuration per GPU; `n8_model1d_test.txt`). Reflection is measured in the L_2 (energy) norm — trapezoid sums of Gaussian pulses converge exponentially, so the deviation column is limited by machine arithmetic, not by measurement or truncation. Analytic L_2 relation: $\|q_{\rm in}\|_{L_2}/\|q_{\rm out}\|_{L_2} = \frac{\mathring{\beta}}{2+\mathring{\beta}}\sqrt{\frac{1+\mathring{\beta}}{1-\mathring{\beta}}}$. Each row sits at its float64 floor: rows 1–3 at the per-sample roundoff floor (deviation saturates between N=8193 and N=16385: 1.9 vs 2.1×10^{-14}); the injection row at the linear roundoff-accumulation optimum ($\sim$ steps $\times\, \varepsilon$; ladder N=16385/32769/65537/131073 $\Rightarrow$ $5.4/2.4/5.9/30 \times 10^{-11}$, non-monotone and insensitive to CFL 0.2 vs 0.05).

boundary condition	measured	exact	deviation
Sommerfeld, L_2 ratio (N=16385)	0.177752646227	0.177752646227	2.1×10^{-14} (float64 floor)
Sommerfeld, L_2 ratio (N=8193)	—	—	1.9×10^{-14} (saturated)
$(\partial_t + c_+\partial_x)$ [P1/CCM op., N=16385]	$R_{L_2} = 1.9 \times 10^{-14}$	0	machine precision
CCM injection (23), rel. L_2 (N=32769)	2.4×10^{-11}	0	roundoff-accumulation optimum

I. Dynamical test at the parameters of Ref. [3] (nodes N10–N11)

Full 3+1 evolution: AthenaK Z4c [4] (vendored, pinned; RK4, CFL 0.25, sixth-order stencils at `nghost` = 4, $\kappa_1 = 0.02$), with (20) realized at the Cauchy outer boundary as described below. This subsection is self-contained: every symbol used by the test is defined here or in Secs. I A–I E.

a. Exact solution injected. Even-parity $(\ell, m) = (2, 0)$ Teukolsky wave [5] with the Gaussian profile of Ref. [3], written with physicists' Hermite polynomials H_n and the origin-regular retarded/advanced combinations:

$$F(u) = X\, e^{-(u-r_c)^2/\tau^2}, \qquad F^{(n)}(u) = X\left(-\frac{1}{\tau}\right)^n H_n\!\left(\frac{u-r_c}{\tau}\right) e^{-(u-r_c)^2/\tau^2}, \tag{27}$$

$$Q^{(n)}(t,r) = F^{(n)}(t-r) - (-1)^n F^{(n)}(t+r), \qquad R^{(n)}(t,r) = F^{(n)}(t-r) + (-1)^n F^{(n)}(t+r), \qquad \partial_t Q^{(n)} = R^{(n+1)}. \tag{28}$$

Radial functions (regular at $r = 0$ by (28); calligraphic to keep A for the boost (14)):

$$\mathcal{A} = 3\Big[\frac{Q^{(2)}}{r^3} + \frac{3Q^{(1)}}{r^4} + \frac{3Q^{(0)}}{r^5}\Big], \qquad \mathcal{B} = -\Big[\frac{Q^{(3)}}{r^2} + \frac{3Q^{(2)}}{r^3} + \frac{6Q^{(1)}}{r^4} + \frac{6Q^{(0)}}{r^5}\Big],$$
$$\mathcal{C} = \frac{1}{4}\Big[\frac{Q^{(4)}}{r} + \frac{2Q^{(3)}}{r^2} + \frac{9Q^{(2)}}{r^3} + \frac{21Q^{(1)}}{r^4} + \frac{21Q^{(0)}}{r^5}\Big], \tag{29}$$

orthonormal-frame metric perturbation ($\dot h_{ij}$ from $Q^{(n)} \to R^{(n+1)}$ termwise):

$$h_{\hat r\hat r} = \mathcal{A}\,\frac{1+3\cos 2\theta}{2}, \quad h_{\hat r\hat\theta} = -3\mathcal{B}\sin\theta\cos\theta, \quad h_{\hat\theta\hat\theta} = 3\mathcal{C}\sin^2\theta - \mathcal{A}, \quad h_{\hat\phi\hat\phi} = -3\mathcal{C}\sin^2\theta + \mathcal{A}\,(3\sin^2\theta - 1), \tag{30}$$

$$\gamma_{ij}(0,x) = \delta_{ij} + h_{ij}(0,x), \qquad K_{ij}(0,x) = -\tfrac{1}{2}\,\dot h_{ij}(0,x) \qquad \text{(linear order; Ref. [3] solves XCTS).} \tag{31}$$

Weyl scalars of the solution at linear order (the self-datum and the extraction reference):

$$r^5\,\psi_0(t,r,\theta) = -\sqrt{\tfrac{27\pi}{10}}\;F^{(2)}(t-r)\;{}_2Y_{20}(\theta), \qquad {}_2Y_{20} = \tfrac{1}{4}\sqrt{\tfrac{15}{2\pi}}\,\sin^2\theta, \qquad r\,\psi_4\big|_{(2,0)} \;\to\; -\sqrt{\tfrac{6\pi}{5}}\,F^{(6)}(t-r). \tag{32}$$

b. Discrete realization of (20). At every Cauchy-boundary point, after the CPBC/gauge rows (21)–(22) (Sommerfeld task of Ref. [4]), the two physical rows are driven through $\tilde A_{ij}$:

$$s^a = \frac{x^a/r}{\sqrt{\gamma_{bc}\,x^b x^c}/r}, \qquad e_\phi \propto \hat z \times s, \quad e_\theta = s \times e_\phi, \qquad \bar m = \tfrac{1}{\sqrt 2}\,(e_\theta - i\,e_\phi), \tag{33}$$

$$\boxed{\partial_t \tilde A_{ab}\big|_{\partial\Omega} \;\mathrel{+}=\; -\frac{2\chi}{\alpha}\left(\psi_0'\,\bar m_a \bar m_b + \bar\psi_0'\,m_a m_b\right), \qquad \psi_0' = A^2\,\psi_0^{\rm datum}, \quad A = \left(\alpha - \gamma_{ij}\beta^i s^j\right)e^{-2\hat\beta}} \tag{34}$$

with $\psi_0^{\rm datum}$ the right-hand member of (32) evaluated at the boundary point ($\hat\beta = 0$ on the flat background); $\psi_0^{\rm datum} \to 0$ reduces (34) to the unmodified scheme exactly (V6; machine-zero reduction in Table II methodology, AthenaK battery: Minkowski preserved to 2.6×10^{-33}).

c. Exact finite-radius reference (node N12). The extraction reference is the *analytic* waveform at the extraction radius itself (no auxiliary reference run): in the AthenaK extraction convention (= $-1\times$ the scri member of (32) continued to finite r),

$$r\,\psi_4^{(2,0)}(t,r) = \sqrt{\tfrac{6\pi}{5}}\Big[F^{(6)} + \frac{2F^{(5)}}{r} + \frac{3F^{(4)}}{r^2} + \frac{3F^{(3)}}{r^3} + \frac{3F^{(2)}}{2\,r^4}\Big](t-r) \;-\; \sqrt{\tfrac{6\pi}{5}}\,\frac{3F^{(2)}}{2\,r^4}(t+r), \tag{35}$$

verified by seven symbolic gates (tracelessness, coefficient-exact linearized vacuum, $\sin^2\theta$ factorization, peeling, scri limit, independent 30-digit finite-difference cross-check at 10^{-40}) and — independently — by the SpECTRE `CharacteristicExtract` cross oracle below.

d. Protocol and result. Parameters of Ref. [3]: $r_c = 20$, $\tau = 2$, $X = 10^{-5}$, Cauchy boundary (worldtube analog) at 41, $t \in [0, 80]$; extraction of $r\psi_4^{(2,0)}$ at $r = 36$; resolution ladder $h = 0.5,\ 0.331,\ 0.277$ (4×A100, every run $< 600\,$s). Error against (35) at each rung's own output times:

$$E(h) = \frac{\big\|r\psi_4^{(2,0)}[h] - r\psi_4^{(2,0)}[\text{exact}]\big\|_2}{\big\|r\psi_4^{(2,0)}[\text{exact}]\big\|_2} = 4.99\times10^{-2},\quad 5.36\times10^{-3},\quad 2.02\times10^{-3}, \tag{36}$$

with measured convergence order 5.39 (first pair) and 5.52 (second pair), finest-rung amplitude $1 \pm 2\times10^{-4}$ and time shift 0.000 (Fig. 1). Two production defects were found *only* by this exact-reference program and are corrected in all data shown: a frame-orientation error in the initial data ($\hat{e}_\theta$ flipped; linearized vacuum violated at 5% of $|R_{ij}|$, invisible to constraint monitors and to run-vs-reference protocols since it sat on both sides) and a one-step offset in the waveform time labels. The self-datum CCM−Sommerfeld difference remains peeling-suppressed ($\psi_0 \sim r^{-5}$, relative imprint $\sim 10^{-9}$ of peak at this worldtube), as in Ref. [3] Sec. V.A.

J. Live Cauchy–characteristic matching inside AthenaK (nodes N13–N14)

The $\psi_0^{\rm datum}$ provider of (34) is replaced by a *native, in-process* characteristic solver evolving in lockstep with the Cauchy evolution (one process, no file round-trips). On the compactified radial coordinate $y = 1 - 2r_{\rm wt}/r$ (Chebyshev–Gauss–Lobatto collocation, LU-factorized global solves, SAT time advance), the solver evolves the *unringed* ℓ=2 t-scalars $(J_t, Q_t, U_t, W_t, H_t)$ of the linearized Bondi–Sachs hierarchy with the β sources required by live (worldtube-anchored gauge) data — machine-derived as exact-rational fits on the anchored and plain solution families jointly:

$$\begin{aligned}
&\partial_r(r^2Q) = -\,r^2\,\partial_r(\bar\eth J) - 4r\,\eth\beta, && \partial_r U = Q/r^2,\\
&\partial_r(r^2W) = \tfrac12\bar\eth\bar\eth J + 2r\,\bar\eth U + \tfrac{r^2}{2}\partial_r(\bar\eth U) - \tfrac43\,\bar\eth\eth\beta, && \partial_r(rH) = \tfrac12 r\,\partial_r^2 J + \partial_r J + \frac{\eth^2\beta}{r} - \frac{\eth Q}{2r} - \eth U,
\end{aligned} \tag{37}$$

with $H = \partial_u J$ and β r-independent at this order. The ringed variables $(rJ, r^2U, \dots)$ diverge at scri on anchored-gauge data ($k = 0, 1$ tails); the unringed system (37) is regular on $y \in [-1, 1]$ in both gauges, and $\psi_0^t = -\partial_r J_t/(2r) - \partial_r^2 J_t/4$ is gauge-invariant (the anchored tails are exactly ψ_0-silent). Worldtube data are sampled live from the Cauchy ADM state on three geodesic spheres (ℓ=2 projections; $\partial_t h = -2K$ at linear order) and mapped by a six-row local contract to the anchored boundary scalars $(J, Q, U, W, H, \beta)_{\rm wt}$. Cone labeling is retarded, $u = t - r$: the boundary datum at Cauchy time t is the ψ_0 probe at r_B on the earlier cone $u = t - r_B$ (causality lag, served from a per-substep history).

Verification. (i) Standalone: anchored end-to-end ψ_0 probe within 8.8×10^{-5} of the exact datum (7.1×10^{-6} at halved step; truncation-dominated). (ii) Live analytic-stub battery: mode-(solver) vs. mode-(self-datum) waveforms agree to 4.1×10^{-9} of peak. (iii) For the purely outgoing test the live datum is conditioning-limited *by physics* ($\psi_0/J \sim 10^{-8}$ peeling: fidelity would need 10^{-11}-relative tube data); the fidelity demonstration instead uses the characteristic-pulse-injection test of Ref. [3] Sec. V.C (ingoing ℓ=2 pulse on the initial null slice, quiescent Cauchy data, $\psi_0/J = 7\times10^{-3}$): the live datum matches the verified solver to 7×10^{-8} of peak in the causally clean window; the transmitted burst exits the Cauchy domain at $t_2 = t_1 + 2R$ within 0.12; a no-datum control shows the entire response is datum-attributable (2.9×10^{-10} of peak) [Fig. 2(a,b)]. (iv) Cross-oracle: SpECTRE `CharacteristicExtract` (the CCE of Ref. [3]), run on a Bondi–Sachs worldtube file generated from the exact solution, reproduces $\Psi_4 = -\sqrt{6\pi/5}\,F^{(6)}$ to 6×10^{-6} relative and the datum normalization $\Psi_0 = -\frac98 F^{(2)}$ (s^2 shape) to 0.1% [Fig. 2(c)]. (v) Constraint rows: a frozen-coefficient LF/GKS analysis of the (Θ, Z_s) boundary realizations shows every matched-falloff advection closure is stable and already absorbs the constraint pair at $1 - O\big((\sigma/2\omega r)^2\big)$; falloff-mismatched sink rows self-generate $O(10\%)$ pair reflection — the production default remains the matched (stock) realization of the (21) rows.

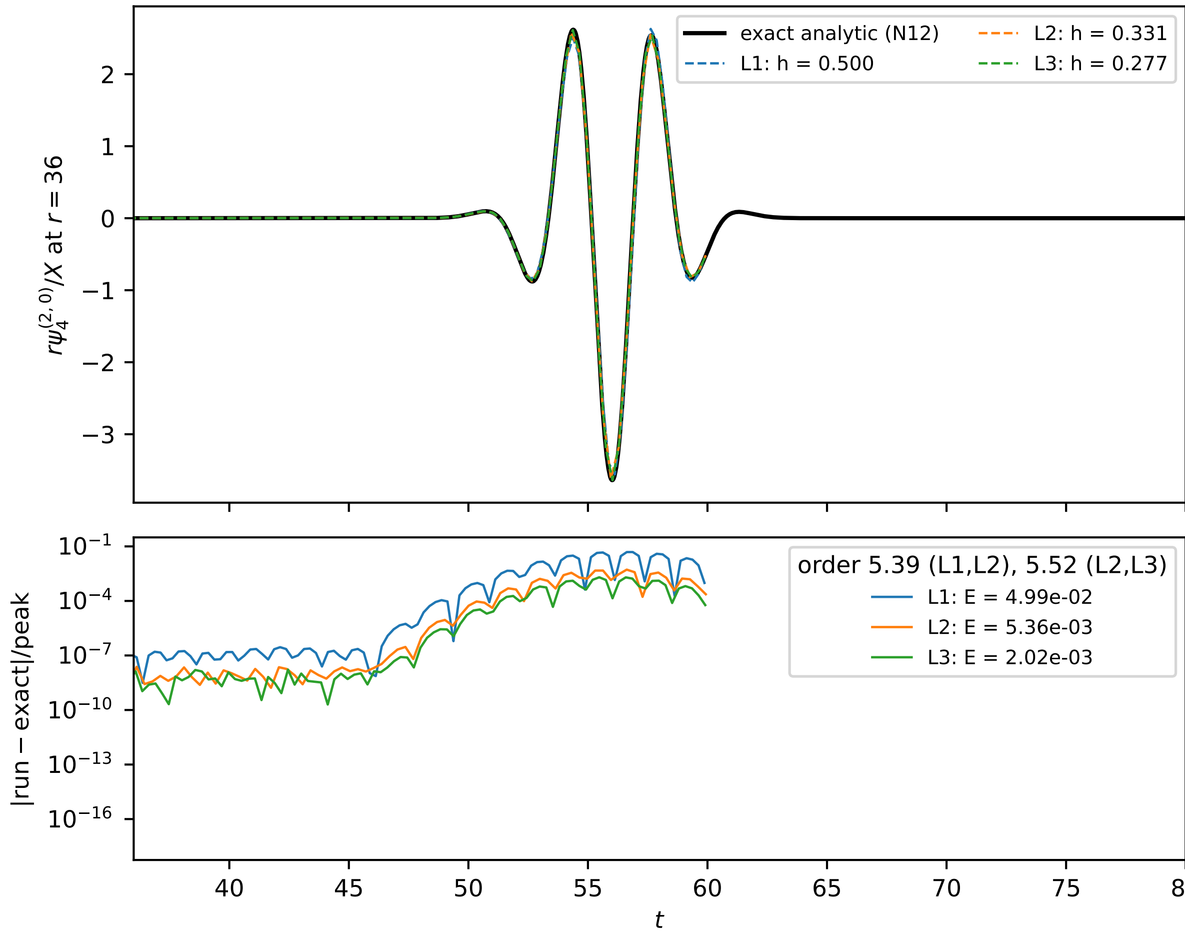

Figure 1. Exact-reference Teukolsky test of Sec. II (r_c=20, τ=2, boundary 41, extraction r=36, X=10^{-5}). Top: (ℓ, m)=$(2, 0)$ mode of $r\psi_4$ — the exact finite-radius waveform (35) (solid) and the three-rung sixth-order ladder. Bottom: pointwise errors against the exact reference; measured order 5.39/5.52, finest-rung relative L_2 error 2.02×10^{-3} (36). Generator: `scripts/plot_paper_figures_v2.py`; data: `results/numerical/athenak_teuk_ana/gpu6/` (ledger node N12).

[1] M. Ruiz, D. Hilditch, and S. Bernuzzi, Constraint preserving boundary conditions for the Z4c formulation of general relativity (2010), local source: `ref-paper/arxiv-1010.0523v2/`, arXiv:1010.0523 [gr-qc].

[2] J. Moxon, M. A. Scheel, and S. A. Teukolsky, Improved Cauchy-characteristic evolution system for high-precision numerical relativity waveforms (2020), local source: `ref-paper/arxiv-2007.01339/`, arXiv:2007.01339 [gr-qc].

[3] S. Ma *et al.*, Fully relativistic three-dimensional Cauchy-characteristic matching for physical degrees of freedom (2023), local source: `ref-paper/arxiv-2308.10361/`, arXiv:2308.10361 [gr-qc].

[4] H. Zhu *et al.*, Performance-portable numerical relativity with AthenaK (2024), local source: `ref-paper/arxiv-2409.10383/`; vendored fork pinned in `athenak/PROVENANCE.md`, arXiv:2409.10383 [gr-qc].

[5] S. A. Teukolsky, Phys. Rev. D **26**, 745 (1982).

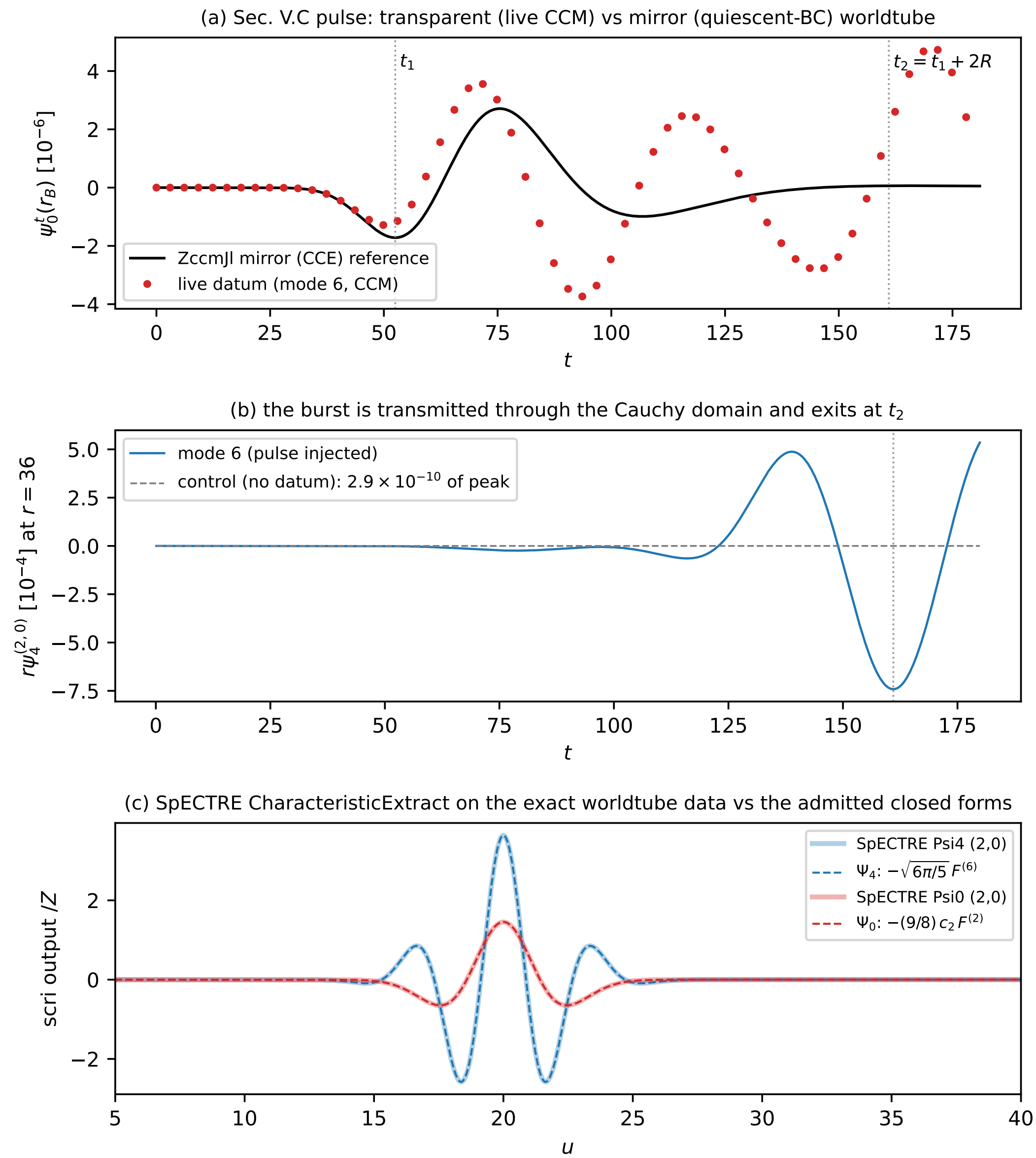

Figure 2. Live-CCM results (Sec. I J). (a) The characteristic-pulse fidelity test: live ψ_0 datum at the Cauchy boundary vs. the quiescent-worldtube (mirror/CCE) reference — the early-window agreement is 7×10^{-8} of peak; the later $O(1)$ separation *is* the matching physics (the live run transmits and later re-receives the pulse). (b) The transmitted burst at the extraction radius exits at $t_2 = t_1 + 2R$; the no-datum control is consistent with zero. (c) SpECTRE `CharacteristicExtract` on the exact worldtube data vs. the closed forms (shaded wide: oracle output; dashed: $-\sqrt{6\pi/5}F^{(6)}$ and $-\frac{9}{8}c_2F^{(2)}$). Generator: `scripts/plot_paper_figures_v2.py`; data: `results/numerical/n14_native/`, `n14_pulse_ref.csv`, `spectre_oracle/` (ledger nodes N13–N14).